SIMBLOCKS // RESULT

2026-10-15 22:35Z

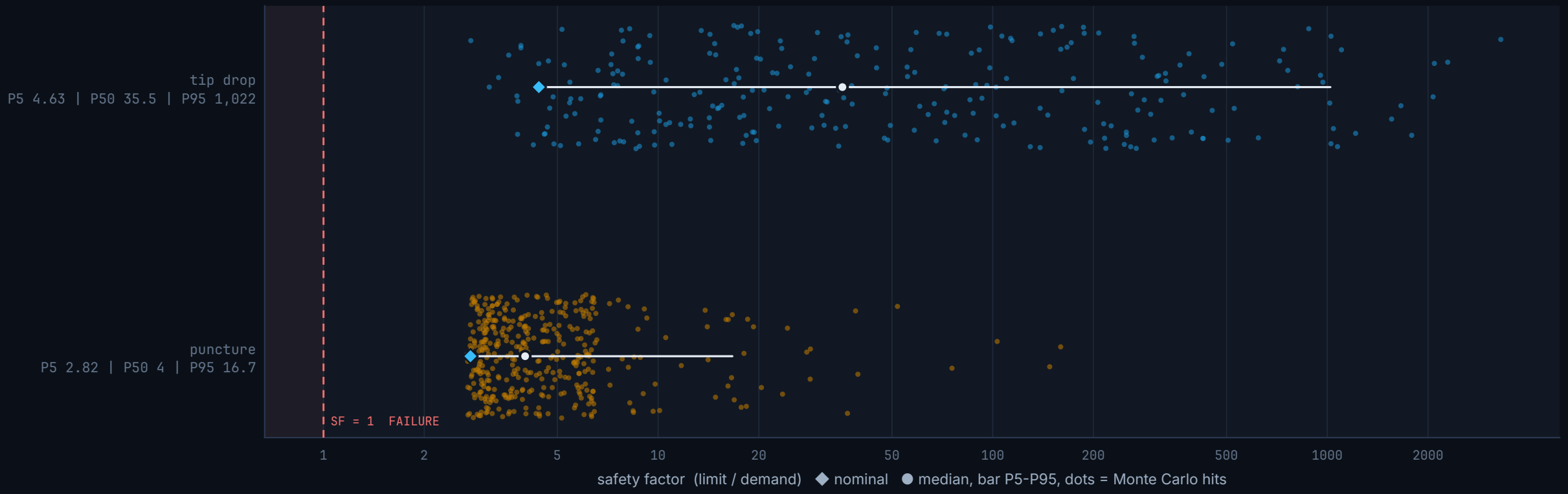
WEDGE 2.5 MM, HORIZONTAL STRIKE, CAD GEOMETRY

Does a 2.5 mm 4130 wedge survive a worst-allowable 600 J horizontal hit?

● PASS VERIFIED

IS 2.5 MM OK?	TIP DROP	PUNCTURE [GOVERNING]
PASS	PASS	PASS
YES	SF 4.4	SF 2.75
worst tip drop 1.14 / 5 mm (geometry max 5.35)	P(fail) 0% | SF P5 5.24	P(fail) 0% | SF P5 2.82

Margin to each user-defined failure mode

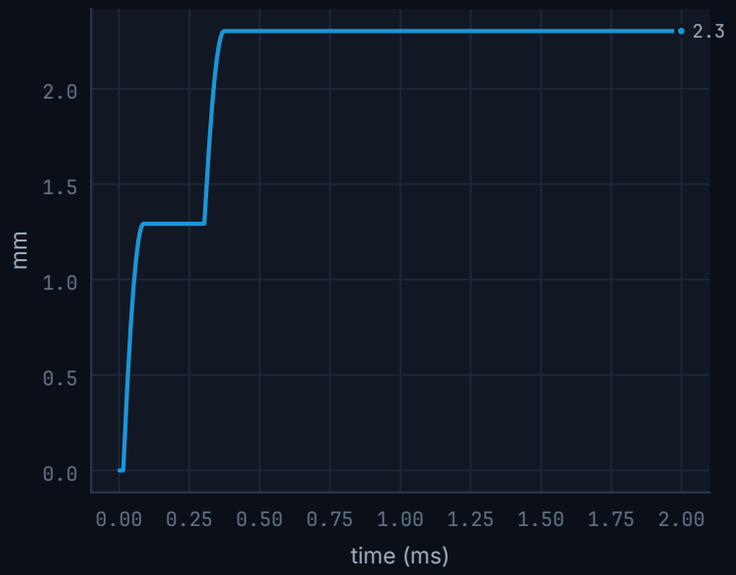


Wedge 2.5 mm, horizontal strike, cad geometry: VERIFIED | PASS 12 WARN 0 FAIL 0 SKIP 0 | 415 runs, 5.3 s

WEDGE 2.5 MM, HORIZONTAL STRIKE, CAD GEOMETRY

Time histories of the governing states (nominal case)

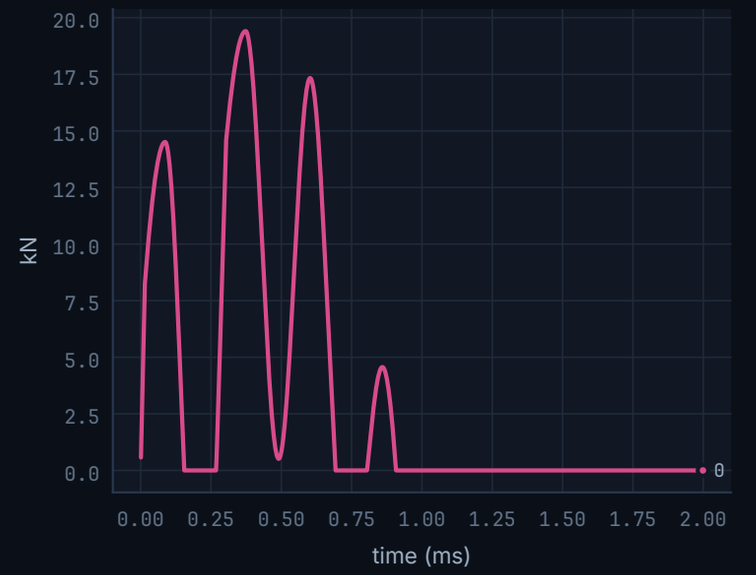
Permanent set at the lip



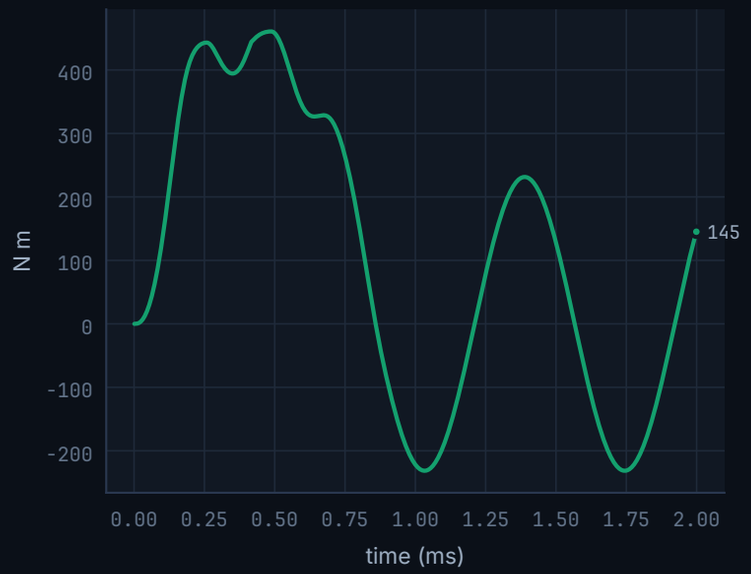
Root hinge rotation set



Tooth force



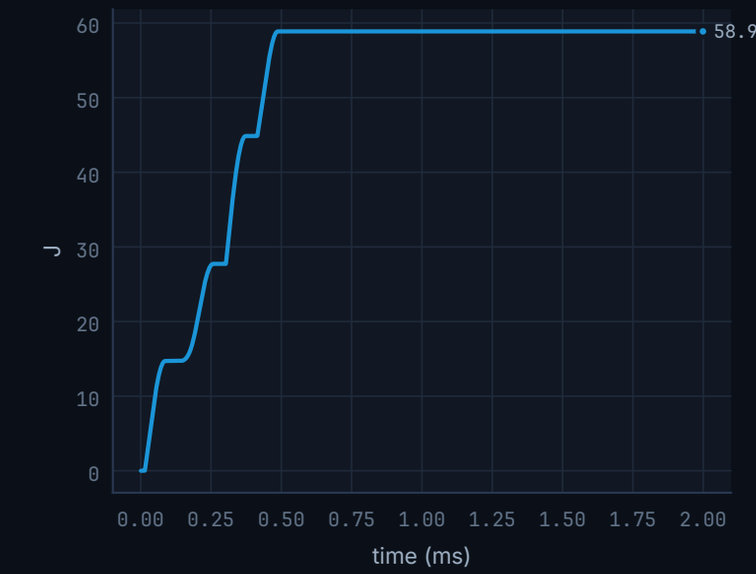
Root moment



Weapon energy



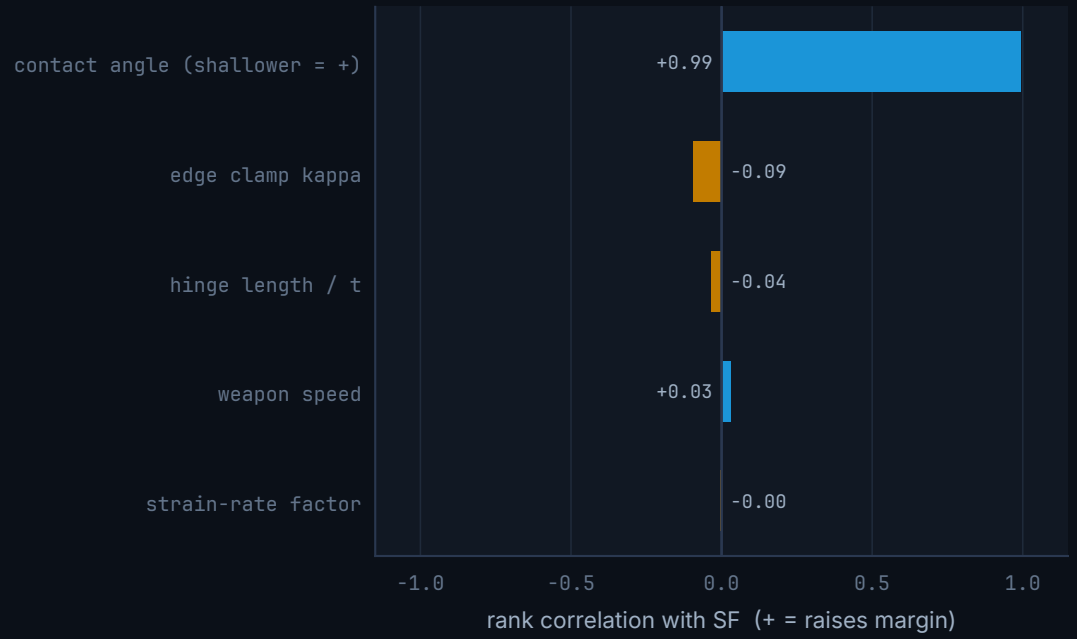
Plastic work absorbed



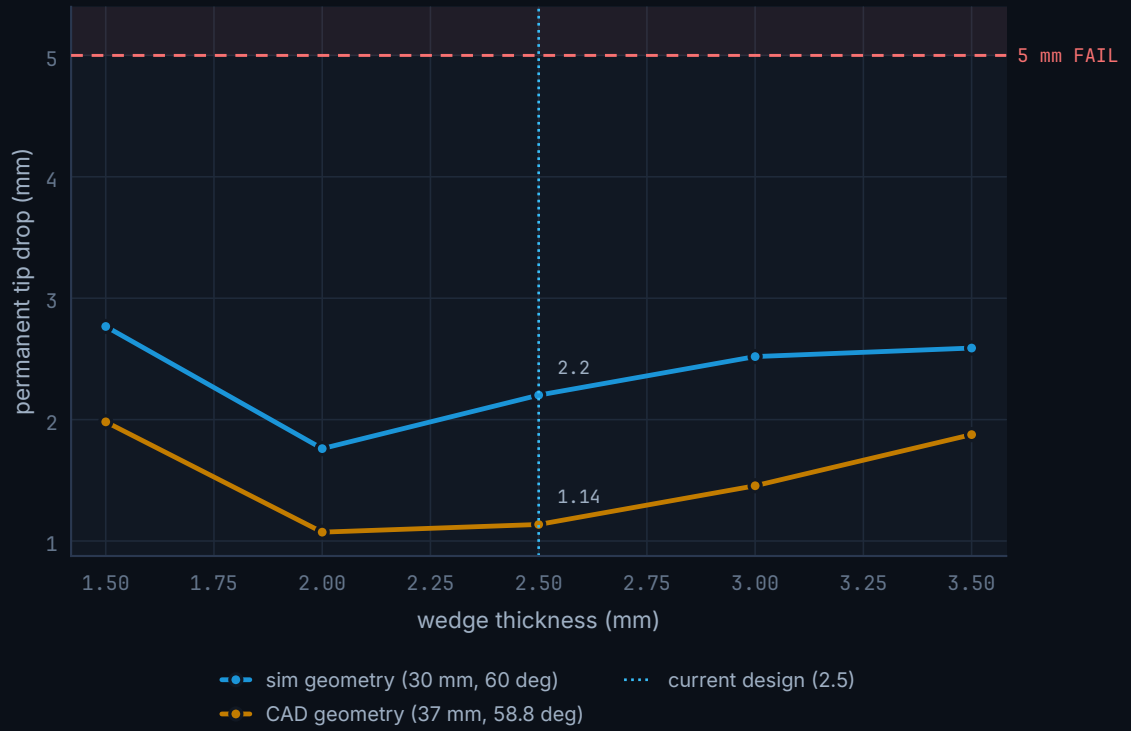
WEDGE 2.5 MM, HORIZONTAL STRIKE, CAD GEOMETRY

What to design around: drivers of the governing margin and the design-variable sweep

Drivers of 'puncture' margin



Tip drop vs thickness at the worst-allowable hit



SIMBLOCKS // VERIFICATION

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WEDGE 2.5 MM, HORIZONTAL STRIKE, CAD GEOMETRY

● PASS VERIFIED

Automated verification battery

CHECK	STATUS	DETAIL
determinism	● PASS	bit-identical repeat
finite_outputs	● PASS	all outputs finite
physical_bounds	● PASS	2 bounds respected
stability_margin	● PASS	critical dt 6.95e-05 s / dt 1e-07 s = 695.0x
dt_convergence	● PASS	tip_drop_perm_mm: err 1e-05% (order 1.2); bending deformation: err 1.1e-05% (order 1.2); denting deformation: err 5.3e-05% (order 1.6); peak_dent_f...
energy_ledger	● PASS	worst closure residual 2% across 3 dt levels (tol 5%)
limit:weapon stopped → no damage	● PASS	matches expected limit
knockout:no dent plasticity	● PASS	tip_drop_perm_mm +154%, bending deformation +184%, denting deformation -100%, peak_dent_force_N +73.2%
knockout:no bend plasticity	● PASS	tip_drop_perm_mm -100%, bending deformation -100%, denting deformation +11.1%, peak_dent_force_N +6.4%
sensitivity	● PASS	5 inputs swept low/high (trends reported, not judged)
monte_carlo	● PASS	400 samples, 0 failed finite/bounds/ledger (0.0%)
failure_modes	● PASS	tip drop: nominal SF 4.4, P(fail) 0%; puncture: nominal SF 2.75, P(fail) 0%